Calculus II

Tutorial Problems #4 10–14 Mar 2025

Topics:

- Gradient vectors
- Tangent planes and differentials
- Linear approximations
- Directional derivatives

Problem 1. Find the gradient of the function at the given point. Then sketch the gradient together with the level curve that passes through the point.

$$f(x, y) = \sqrt{2x + 3y}, \quad (-1, 2).$$

Problem 2. Find ∇f at the given point

$$f(x, y, z) = (x^2 + y^2 + z^2)^{-1/2} + \ln(xyz), \quad (-1, 2, -2).$$

Problem 3. Find the derivative of the function at P_0 in the direction of $\mathbf{u}$.

$$g(x, y) = \frac{x - y}{xy + 2}, \quad P_0(1, -1), \quad \mathbf{u} = 12\mathbf{i} + 5\mathbf{j}.$$

Problem 4. Find the directions in which the functions increase and decrease most rapidly at P_0. Then find the derivatives of the functions in these directions.

$$f(x, y) = x^2 + xy + y^2, \quad P_0(-1, 1).$$

Problem 5. Sketch the curve $f(x, y) = c$ together with ∇f and the tangent line at the given point. Then write an equation for the tangent line.

$$x^2 + y^2 = 4, \quad (\sqrt{2}, \sqrt{2}).$$

Problem 6. Let $f(x, y) = \frac{(x-y)}{(x+y)}$. Find the directions $\mathbf{u}$ and the values of $D_{\mathbf{u}} f\left(-\frac{1}{2}, \frac{3}{2}\right)$ for which

a. $D_{\mathbf{u}} f\left(-\frac{1}{2}, \frac{3}{2}\right)$ is largest.

b. $D_{\mathbf{u}}f\left(-\frac{1}{2},\frac{3}{2}\right)$ is smallest.

c. $D_{\mathbf{u}}f\left(-\frac{1}{2},\frac{3}{2}\right)=0$.

Problem 7. Is there a direction $\mathbf{u}$ in which the rate of change of $f(x,y)=x^2-3xy+4y^2$ at $P(1,2)$ equals to 14? Justify your answer.

Problem 8. Find parametric equations for the line tangent to the curve of intersection of the surfaces at the given point.

(a) Surfaces: $x+y^2+2z=4, \quad x=1.$
Point: $(1,1,1)$.

(b) Surfaces:

$$x^3+3x^2y^2+y^3+4xy-z^2=0,$$
$$x^2+y^2+z^2=11$$

Point: $(1,1,3)$.

Problem 9. By about how much will

$$g(x,y,z)=x+x\cos z-y\sin z+y$$

change if the point $P(x,y,z)$ moves from $P_0(2,-1,0)$ a distance of $ds=0.2$ unit toward the point $P_1(0,1,2)$.

Problem 10. Find the linearization $L(x,y)$ of the function at each point.

$$f(x,y)=x^2+y^2+1$$

at **a.** $(0,0)$, **b.** $(1,1)$.

END

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Answers to selected problems

1. $\left(\frac{1}{2},\frac{3}{4}\right)$
2. $\left(-\frac{26}{27},\frac{23}{54},-\frac{23}{54}\right)$
3. $\frac{21}{13}$
4. $\sqrt{2},\ -\sqrt{2}$
5. $x+y-2\sqrt{2}=0$

6. (a) $\left(\frac{3}{\sqrt{10}}, \frac{1}{\sqrt{10}}\right)$ (b) $-\left(\frac{3}{\sqrt{10}}, \frac{1}{\sqrt{10}}\right)$ (c) $\pm\left(-\frac{1}{\sqrt{10}}, \frac{3}{\sqrt{10}}\right)$.
7. does not exist
8. $x = 1,\ y = 1 + 2t,\ z = 1 - 2t$
9. 0
10. (a) 1 (b) $2x + 2y - 1$.